

ENGR 213 · Applied Ordinary Differential Equations

Lecture 6: Differential Equations as Mathematical Models & Linear Models (Explained)

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Executive Overview & Lecture Roadmap

Lecture 6 marks the fundamental bridge between purely formal ODE integration techniques and real-world engineering systems. When describing physical phenomena, empirical principles and conservation laws are formulated as **rates of change** ($\frac{dy}{dt}$), naturally producing differential equations.

Phenomenon	Governing differential equation	Solution method
Population growth	$\frac{dP}{dt} = kP \ (k > 0)$	Separable
Radioactive decay	$\frac{dA}{dt} = kA \ (k < 0)$	Separable
Newton's cooling/warming	$\frac{dT}{dt} = k(T - T_m) \ (k < 0)$	Separable or linear
Single-tank mixture	$\frac{dA}{dt} = R_{in} - R_{out}$	First-order linear
LR series circuit	$L \frac{di}{dt} + Ri = E(t)$	First-order linear
RC series circuit	$R \frac{dq}{dt} + \frac{1}{C}q = E(t)$	First-order linear

1. Steps in the Mathematical Modeling Process

Building a reliable engineering model follows five systematic stages:

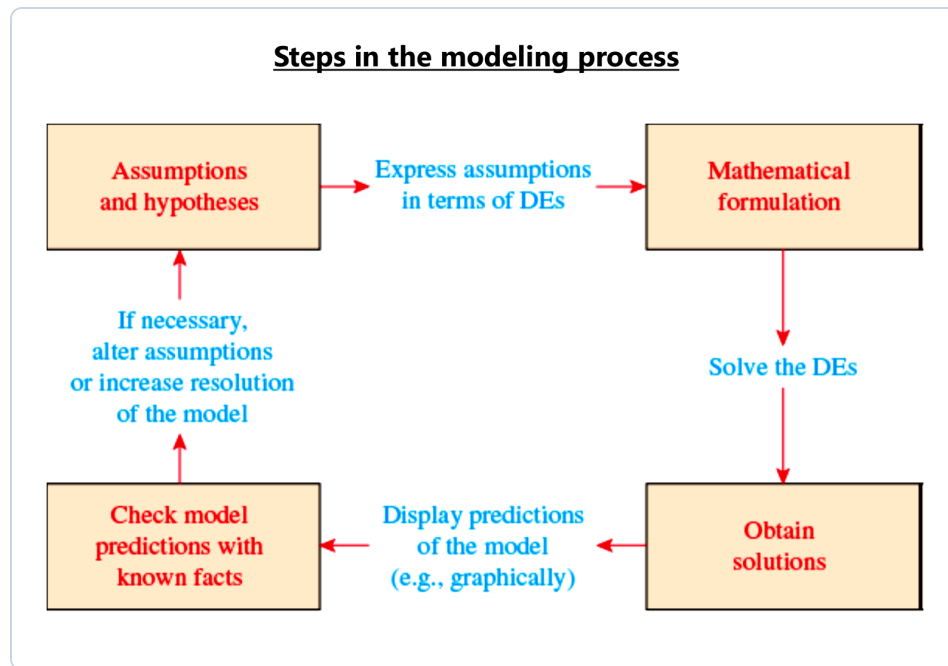


Figure 1: Steps in the modeling process, Lecture 6, slide 4 (Zill Fig. 1.3.1).

Reading the cycle, clockwise from the top left:

1. **Assumptions and hypotheses** about the real system (which variables matter, which effects to ignore).
2. **Express the assumptions in terms of DEs**, producing the **mathematical formulation**.
3. **Solve the DEs** to **obtain solutions**.
4. **Display predictions of the model** (for example graphically).
5. **Check the model predictions with known facts** (experimental data).
6. **If necessary, alter the assumptions or increase the resolution of the model**, and go around again.

The loop is the point: a model is never "done" after one pass. If the predictions compare poorly with data, you refine the assumptions, usually at the price of more mathematical complexity.

2. Exponential Growth & Radioactive Decay

2.1 Theoretical Derivation

- **Hypothesis:** The rate of change of a quantity is directly proportional to the amount currently present:

$$\frac{dP}{dt} \propto P \implies \frac{dP}{dt} = kP$$

where k is the constant of proportionality.

- If $k > 0$: Exponential growth (microbiology, unrestricted population, compound interest).
- If $k < 0$: Exponential decay (nuclear physics, radioisotope dating, drug pharmacokinetics).
- **Integration:**

$$\frac{dP}{P} = k dt \implies \ln |P| = kt + C_1 \implies P(t) = P_0 e^{kt}$$

where $P_0 = P(0)$ represents the initial quantity at $t = 0$.

2.2 Lecture 6 — Example 1: Bacterial Growth

- **Problem:** A culture initially has P_0 bacteria. At $t = 1$ h, the number is measured to be $\frac{3}{2}P_0$. Assuming growth is proportional to the current population, determine the time necessary for the population to triple.
- **Step 1: Formulate the Model & General Solution:**

$$\frac{dP}{dt} = kP \implies P(t) = P_0 e^{kt}$$

- **Step 2: Determine Rate Constant k from Benchmark Data:**

$$P(1) = \frac{3}{2}P_0 \implies P_0 e^{k(1)} = \frac{3}{2}P_0 \implies e^k = 1.5$$

$$k = \ln(1.5) = \ln\left(\frac{3}{2}\right) \approx 0.405465 \text{ h}^{-1}$$

- **Step 3: Solve for Tripling Condition $P(t) = 3P_0$:**

$$P_0 e^{kt} = 3P_0 \implies e^{kt} = 3 \implies kt = \ln 3$$

$$t = \frac{\ln 3}{k} = \frac{\ln 3}{\ln(1.5)} = \frac{1.098612}{0.405465} \approx 2.7095 \text{ hours}$$

- **Step 4: Engineering Conversion:**

$$t \approx 2 \text{ hours and } (0.7095 \times 60) \text{ min} \approx 2 \text{ h } 43 \text{ min}$$

3. Newton's Law of Cooling and Warming

3.1 Physical Principles & Governing Equation

Newton's empirical law states that the rate of change of temperature $\frac{dT}{dt}$ of a body is proportional to the instantaneous temperature difference between the body $T(t)$ and the surrounding medium T_m :

$$\frac{dT}{dt} = k(T - T_m)$$

□ Newton's Law of Cooling/Warming

- Newton's empirical law of cooling (or warming): the rate at which the temperature of a body changes (dT/dt) is proportional to the difference between the temperature of the body ($T(t)$) and the temperature of the surrounding medium (T_m):

$$\frac{dT}{dt} \propto (T - T_m) \longrightarrow \frac{dT}{dt} = k(T - T_m) \quad (k \text{ is a constant of proportionality})$$

Figure 2: Newton's law of cooling/warming, Lecture 6, slide 8.

Case	Sign of $T - T_m$	Direction of change	So k must be
Object hotter than the room ($T > T_m$)	+	Cools: $dT/dt < 0$	negative
Object colder than the room ($T < T_m$)	−	Warms: $dT/dt > 0$	negative

In both cases $k < 0$: the rate is proportional to the temperature **difference**, and the temperature always moves toward T_m .

3.2 Analytic Solution

$$\frac{dT}{T - T_m} = k dt \implies \ln |T - T_m| = kt + C_1 \implies T(t) - T_m = ce^{kt}$$

$$T(t) = T_m + ce^{kt}$$

Using the initial temperature $T(0) = T_0 \implies c = T_0 - T_m$:

$$T(t) = T_m + (T_0 - T_m)e^{kt}$$

As $t \rightarrow \infty$, since $k < 0$, $e^{kt} \rightarrow 0$, proving that $\lim_{t \rightarrow \infty} T(t) = T_m$.

3.3 Lecture 6 — Example 2: Cooling of a Cake

- **Problem:** When a cake is removed from an oven, its temperature is 300°F . Three minutes later, its temperature is 200°F . The room temperature is 70°F . How long will it take for the cake to cool off to room temperature?
- **Step 1: Set Up Initial Parameters:**
 - Medium temperature: $T_m = 70^{\circ}\text{F}$
 - Initial temperature: $T(0) = 300^{\circ}\text{F}$
 - Temperature difference: $T_0 - T_m = 300 - 70 = 230^{\circ}\text{F}$
 - Governing formula: $T(t) = 70 + 230e^{kt}$
- **Step 2: Solve for Heat Dissipation Constant k :**

$$T(3) = 200 \implies 70 + 230e^{3k} = 200 \implies 230e^{3k} = 130$$

$$e^{3k} = \frac{130}{230} = \frac{13}{23} \implies 3k = \ln\left(\frac{13}{23}\right)$$

$$k = \frac{1}{3} \ln\left(\frac{13}{23}\right) \approx -0.19018 \text{ min}^{-1}$$

- **Step 3: Rigorous Mathematical vs. Engineering Resolution:**
 - **Pure Mathematical Limit:**

$$T(t) = 70 + 230e^{kt} = 70 \implies 230e^{kt} = 0 \implies e^{kt} = 0$$

Since $e^{kt} > 0$ for all finite t , the cake mathematically reaches exactly 70°F only as $t \rightarrow \infty$.

- **Practical Thermal Equilibrium (Engineering Threshold):** In physical thermodynamics, thermal equilibrium is achieved when the temperature difference is imperceptible ($< 0.5^{\circ}\text{F}$):

$$70 + 230e^{kt} = 70.5 \implies e^{kt} = \frac{0.5}{230} = \frac{1}{460}$$

$$t = \frac{\ln(1/460)}{-0.19018} = \frac{-6.13123}{-0.19018} \approx 32.24 \text{ minutes}$$

4. Tank Mixture Dynamics

4.1 Conservation of Mass & General Formulation

The rate of accumulation of solute $A(t)$ (salt, chemical, pollutant) in a thoroughly mixed liquid tank equals input rate minus output rate:

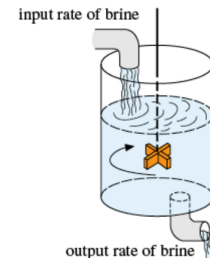
$$\frac{dA}{dt} = R_{in} - R_{out} = c_{in}Q_{in} - c_{out}(t)Q_{out}$$

Where:

- $A(t)$ = mass of solute in the tank at time t [e.g., lb or kg]
- Q_{in}, Q_{out} = volumetric flow rates in and out [e.g., gal/min or L/min]
- c_{in} = concentration of incoming brine [e.g., lb/gal or kg/L]
- $V(t) = V_0 + (Q_{in} - Q_{out})t$ = instantaneous liquid volume in tank
- $c_{out}(t) = \frac{A(t)}{V(t)}$ = instantaneous concentration leaving tank

□ Mixtures

- The mixing of two fluids sometimes gives rise to a linear first-order differential equation.
- For example, consider mixing of two brine solutions of different salt concentrations which gives rise to a first-order DE for the amount of salt contained in the mixture:



$$\frac{dA}{dt} = (\text{input rate of salt}) - (\text{output rate of salt}) \quad * A(t) \text{ denotes the amount of salt in the tank at time } t$$

$$\frac{dA}{dt} = Ri_n - R_{out} \rightarrow \frac{dA}{dt} = c_{\text{salt},in}Q_{in} - c_{\text{salt},out}Q_{out} \quad * Q \text{ denotes flowrate}$$

* C_{salt} is concentration

Different scenarios: $Q_{in} = Q_{out}$ $Q_{in} < Q_{out}$ $Q_{in} > Q_{out}$

Figure 3: Mixtures, Lecture 6, slide 11.

Reading the slide: brine flows in at a rate Q_{in} with concentration c_{in} , the tank is kept well mixed, and the mixture flows out at Q_{out} with the tank's own concentration $c_{out} = A(t)/V(t)$. The net rate of change of salt is **input rate minus output rate**:

$$\frac{dA}{dt} = R_{in} - R_{out} = c_{in}Q_{in} - \frac{A(t)}{V(t)}Q_{out}, \quad V(t) = V_0 + (Q_{in} - Q_{out})t$$

The slide's three scenarios: $Q_{in} = Q_{out}$ (constant volume), $Q_{in} < Q_{out}$ (the tank drains), $Q_{in} > Q_{out}$ (the tank fills).

The universal linear ODE is:

$$\frac{dA}{dt} + \frac{Q_{out}}{V_0 + (Q_{in} - Q_{out})t} A(t) = c_{in} Q_{in}$$

4.2 Lecture 6 — Example 3: Mixture of Two Salt Solutions

- **Problem:** A large tank holds 300 gal of brine. Brine is pumped in at 3 gal/min with salt concentration 2 lb/gal, thoroughly mixed, and pumped out at the same rate of 3 gal/min. The tank initially contains 50 lb of dissolved salt. How much salt will be in the tank after a long time? Find $A(t)$.
- **Step 1: Parameter Identification:**
 - Initial volume: $V_0 = 300$ gal
 - Flow rates: $Q_{in} = 3$ gal/min, $Q_{out} = 3$ gal/min $\implies V(t) = 300$ gal (constant volume).
 - Inflow concentration: $c_{in} = 2$ lb/gal
 - Salt inflow rate: $R_{in} = c_{in} Q_{in} = (2 \text{ lb/gal})(3 \text{ gal/min}) = 6 \text{ lb/min}$.
 - Salt outflow rate: $R_{out} = \left(\frac{A(t)}{300}\right)(3) = \frac{A(t)}{100} \text{ lb/min}$.
 - Initial condition: $A(0) = 50$ lb.
- **Step 2: Construct & Solve Linear ODE:**

$$\frac{dA}{dt} + \frac{1}{100}A = 6$$

- Integrating factor:

$$\mu(t) = e^{\int \frac{1}{100} dt} = e^{t/100}$$

- Multiply and integrate:

$$\frac{d}{dt} [A(t)e^{t/100}] = 6e^{t/100}$$

$$A(t)e^{t/100} = 6 \int e^{t/100} dt = 6(100)e^{t/100} + C = 600e^{t/100} + C$$

$$A(t) = 600 + Ce^{-t/100}$$

- **Step 3: Apply Initial Condition** $A(0) = 50$:

$$50 = 600 + Ce^0 \implies C = 50 - 600 = -550$$

$$A(t) = 600 - 550e^{-t/100} \text{ lb}$$

- **Step 4: Long-Term Asymptotic Behavior** ($t \rightarrow \infty$):

$$\lim_{t \rightarrow \infty} A(t) = 600 - 550(0) = 600 \text{ lb}$$

- **Physical Verification:** The steady-state amount equals total tank volume multiplied by inflow concentration:

$$A_{\infty} = V_0 \times c_{in} = (300 \text{ gal}) \times (2 \text{ lb/gal}) = 600 \text{ lb}$$

5. First-Order Series Electric Circuits

5.1 Kirchhoff's Voltage Law (KVL)

In any closed electrical loop, the impressed electromotive force $E(t)$ equals the sum of voltage drops across circuit components:

$$V_L + V_R + V_C = E(t)$$

Component	Voltage drop	In terms of charge q ($i = dq/dt$)
Inductor L	$L \frac{di}{dt}$	$L \frac{d^2q}{dt^2}$
Resistor R	iR	$R \frac{dq}{dt}$
Capacitor C	$\frac{1}{C}q$	$\frac{1}{C}q$

□ Series Circuits

- For a single-loop LRC-series circuit containing an inductor, resistor, and capacitor, Kirchhoff's second law states: *The impressed voltage $E(t)$ on a closed loop must equal the sum of the voltage drops in the loop.*

$i(t)$: current

$q(t)$: charge on a capacitor

L : inductance (henrys(h))

R : resistance (ohms (Ω))

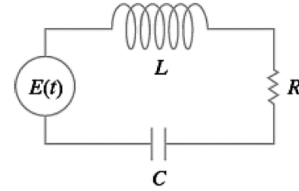
C : capacitance (farads(f))

Voltage drop

Inductor: $L \frac{di}{dt} = L \frac{d^2q}{dt^2}$

Resistor: $iR = R \frac{dq}{dt}$

Capacitor: $\frac{1}{C} q$



LRC-series circuit

$$L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E(t)$$

Figure 4: Series circuits, Lecture 6, slide 15.

Reading the slide: the single loop contains a source $E(t)$, an inductor L , a resistor R and a capacitor C .

Kirchhoff's second law adds the three drops to give the **second-order** equation in the boxed formula, $L q'' + R q' + \frac{1}{C} q = E(t)$. Removing the capacitor leaves the **LR circuit**, $L i' + R i = E(t)$, which is first-order and linear in i : that is why it belongs in this lecture.

5.2 LR-Series Circuit Analysis

When an inductor L and resistor R are connected in series with voltage source $E(t)$:

$$L \frac{di}{dt} + Ri = E(t) \implies \frac{di}{dt} + \frac{R}{L} i = \frac{E(t)}{L}$$

- Integrating factor: $\mu(t) = e^{\int \frac{R}{L} dt} = e^{\frac{R}{L} t}$
- For constant voltage $E(t) = E_0$:

$$i(t) = \frac{E_0}{R} + ce^{-\frac{R}{L} t}$$

If $i(0) = 0$:

$$i(t) = \frac{E_0}{R} \left(1 - e^{-\frac{R}{L} t} \right)$$

- **Steady-State Current:** $i_{ss} = \frac{E_0}{R}$ (inductor acts as a short circuit).
- **Transient Current:** $i_{tr}(t) = -\frac{E_0}{R} e^{-\frac{R}{L} t}$.

- **Inductive Time Constant:** $\tau = \frac{L}{R}$. At $t = \tau$, $i(\tau) = 0.632 i_{ss}$.

5.3 Lecture 6 — Example 4: LR-Series Circuit

- **Problem:** A 12 V battery is connected to an LR series circuit with inductance $L = 0.5 \text{ H}$ and resistance $R = 10 \Omega$. Determine current $i(t)$ if initial current is zero.
- **Step 1: Set Up Governing Equation:**

$$0.5 \frac{di}{dt} + 10i = 12 \implies \frac{di}{dt} + 20i = 24$$

- **Step 2: Solve Using Integrating Factor:**

$$\mu(t) = e^{\int 20 dt} = e^{20t}$$

$$\frac{d}{dt} [ie^{20t}] = 24e^{20t} \implies i(t)e^{20t} = \frac{24}{20}e^{20t} + C = 1.2e^{20t} + C$$

$$i(t) = 1.2 + Ce^{-20t}$$

- **Step 3: Apply Initial Condition $i(0) = 0$:**

$$0 = 1.2 + C \implies C = -1.2$$

$$i(t) = 1.2 (1 - e^{-20t}) \text{ Amperes}$$

- **Step 4: Engineering Interpretation:**
 - Steady-state current: $\lim_{t \rightarrow \infty} i(t) = 1.2 \text{ A}$.
 - Time constant: $\tau = \frac{L}{R} = \frac{0.5}{10} = 0.05 \text{ seconds}$.
 - By $t = 5\tau = 0.25 \text{ seconds}$, current reaches 99.3% of its maximum value.